

# Santiago Alzate

Mathematician | M.A. in Mathematics

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*“The essence of mathematics lies in its freedom.”*

— Georg Cantor (1854–1918)

## AREAS OF INTEREST AND RESEARCH

### Analysis

Real and complex analysis, integration theory, fractional analysis.

### Functional Analysis

Spectral theory and operator theory.

### Algebra and Topology

Category theory, functorial analysis, and non-associative algebra.

## EDUCATION

2024–2026

### M.A. in Mathematics

Binghamton University, New York, USA

Thesis: *The Peter–Weyl Theorem and Its Applications.*

2017–2023

### B.S. in Mathematics

Universidad de Antioquia, Medellín, Colombia

Undergraduate degree in Mathematics.

## TEACHING EXPERIENCE

2026–Present

### Instructor

Institución Universitaria de Envigado, Envigado, Colombia

- Calculus I, Discrete Mathematics, and Logical Reasoning.

2024–2026

### Teaching Assistant

Binghamton University, New York, USA

- Math 226–227** (Techniques of Integral Calculus, Series, and Sequences): Spring 2026.
- Math 224–225** (Differential and Integral Calculus): Fall 2025, Winter 2026, and Summer 2026 (online).
- Math 147** (Elementary Statistics): Fall 2024, Spring 2025, and Summer 2025 (online).

- 2023–Present      **Adjunct Instructor**  
*Institute of Mathematics, Universidad de Antioquia, Medellín, Colombia*
- Differential Calculus, Integral Calculus, and Multivariable Calculus.
  - Vector Geometry and Linear Algebra.
  - **Current semester (2026-2):** Numerical Analysis, Analysis II, and Basic Geometry — advanced courses in the Mathematics program.

## RESEARCH EXPERIENCE

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- 2021–2022      **Undergraduate Researcher**  
*Algebra Research Group, Universidad de Antioquia, Medellín, Colombia*
- Young researcher within the undergraduate thesis project. Research focused on non-associative algebra and category theory.
- SUMMARY** Co-author of 1 research article published in a peer-reviewed journal, and contributor to the solution of 13 mathematical problems published in international journals.

## AWARDS AND SCHOLARSHIPS

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- 2025      **Mian Scholarship in Mathematics**  
*Binghamton University, New York, USA*  
Distinction awarded to outstanding mathematics students during the academic year.
- 2022      **Gilberto García Pulgarín Award**  
*Universidad de Antioquia, Medellín, Colombia*  
Recognition for undergraduate research work in mathematics. Supported by Dr. Rigoberto Flórez, Dr. Fernando Guzmán, and Dr. Leandro Junes.
- 2019      **Gilberto García Pulgarín Award**  
*Universidad de Antioquia, Medellín, Colombia*  
Recognition for outstanding academic and research performance.
- 2017–2023      **Outstanding Student Scholarships**  
*Universidad de Antioquia, Medellín, Colombia*  
Awarded four times for achieving the highest academic average of the semester.

## PUBLICATIONS

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### Peer-reviewed articles

- [1] S. Alzate, O. Correa, R. Flórez, *Fibonacci Identities From Jordan Identities*, *Fibonacci Quarterly*, Vol. 58, No. 5 (2020), pp. 2–20.

### Mathematical problems solved in international journals

- [1] S. Alzate, *La Gaceta de la RSME*, Vol. 25, No. 2 (2022), Problem 419.  
[2] S. Alzate, *La Gaceta de la RSME*, Vol. 24, No. 3 (2021), Problem 408.  
[3] S. Alzate Suárez, *The College Mathematics Journal*, Vol. 52, No. 1 (2021), Problem 1167.  
[4] S. Alzate Suárez, *Fibonacci Quarterly*, Vol. 58, No. 1 (2020), Problem H-819.  
[5] S. Alzate Suárez, *Fibonacci Quarterly*, Vol. 58, No. 2 (2020), Problem B-1246.

- [6] S. Alzate Suárez, *American Mathematical Monthly*, Vol. 127, No. 4 (2020), Problem 12063 (*Limit of nested radicals*).
- [7] S. Alzate Suárez, *The College Mathematics Journal*, Vol. 51, No. 4 (2020), Problem 1157.
- [8] S. Alzate Suárez, *Fibonacci Quarterly*, Vol. 57, No. 1 (2019), Problem B-1222.
- [9] S. Alzate Suárez, *Fibonacci Quarterly*, Vol. 57, No. 2 (2019), Problem B-1226.
- [10] S. Alzate Suárez, *Fibonacci Quarterly*, Vol. 57, No. 2 (2019), Problem B-1227.
- [11] S. Alzate Suárez, *Fibonacci Quarterly*, Vol. 57, No. 3 (2019), Problem B-1231.
- [12] S. Alzate Suárez, *Fibonacci Quarterly*, Vol. 57, No. 3 (2019), Problem B-1233.
- [13] S. Alzate Suárez, *Fibonacci Quarterly*, Vol. 57, No. 3 (2019), Problem B-1235.

## TALKS AND PRESENTATIONS

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|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>Analysis Seminar</b><br><i>Binghamton University, New York, USA</i><br><i>The Peter–Weyl Theorem and Its Applications.</i>                                                               |
| 2024 | <b>Mathematics Seminar</b><br><i>Institute of Mathematics, Universidad de Antioquia, Medellín, Colombia</i><br><i>Solving Mathematical Problems.</i>                                        |
| 2022 | <b>Fifth Meeting on Algebra and Topology</b><br><i>Institute of Mathematics, Universidad de Antioquia, Medellín, Colombia</i><br><i>Some Techniques for Solving Real Analysis Problems.</i> |

## SKILLS AND LANGUAGES

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|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Computer Skills | <b>Wolfram Mathematica</b> (Intermediate), <b>LaTeX</b> (Advanced), <b>MATLAB</b> (Basic), <b>Python</b> (Basic). |
| Languages       | <b>Spanish</b> (Native), <b>English</b> (Advanced / Professional and academic proficiency).                       |